CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 1: Real Numbers

Step-by-step solutions to the textbook exercises: 10 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 1.1

Q1 Break each number into prime factors: (i) 140 (ii) 156 (iii) 3825 (iv) 5005 (v) 7429.

SOLUTION

Keep dividing by the smallest prime that divides the number.

(i) $140 = 2 \times 70 = 2 \times 2 \times 35 = 2^2 \times 5 \times 7$

(ii) $156 = 2 \times 78 = 2 \times 2 \times 39 = 2^2 \times 3 \times 13$

(iii) $3825 = 3 \times 1275 = 3 \times 3 \times 425 = 3^2 \times 5 \times 85 = 3^2 \times 5^2 \times 17$

(iv) $5005 = 5 \times 1001 = 5 \times 7 \times 143 = 5 \times 7 \times 11 \times 13$

(v) $7429 = 17 \times 437 = 17 \times 19 \times 23$

Answer: (i) $2^2 \times 5 \times 7$ **(ii)** $2^2 \times 3 \times 13$ **(iii)** $3^2 \times 5^2 \times 17$ **(iv)** $5 \times 7 \times 11 \times 13$ **(v)** $17 \times 19 \times 23$

Q2 For each pair, find the HCF and LCM, and check that HCF $\times$ LCM equals the product of the pair: (i) 26 and 91 (ii) 510 and 92 (iii) 336 and 54.

SOLUTION

HCF is the product of the smallest powers of the common primes. LCM is the product of the greatest powers of all the primes.

(i) $26 = 2 \times 13$ and $91 = 7 \times 13$. HCF $= 13$ and LCM $= 2 \times 7 \times 13 = 182$.

Check: $13 \times 182 = 2366$ and $26 \times 91 = 2366$.

(ii) $510 = 2 \times 3 \times 5 \times 17$ and $92 = 2^2 \times 23$. HCF $= 2$ and LCM $= 2^2 \times 3 \times 5 \times 17 \times 23 = 23460$.

Check: $2 \times 23460 = 46920$ and $510 \times 92 = 46920$.

(iii) $336 = 2^4 \times 3 \times 7$ and $54 = 2 \times 3^3$. HCF $= 2 \times 3 = 6$ and LCM $= 2^4 \times 3^3 \times 7 = 3024$.

Check: $6 \times 3024 = 18144$ and $336 \times 54 = 18144$.

Answer: (i) HCF 13, LCM 182 (ii) HCF 2, LCM 23460 (iii) HCF 6, LCM 3024. Verified in each case.

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Q3 Find the LCM and HCF by the prime factorisation method: (i) 12, 15 and 21 (ii) 17, 23 and 29 (iii) 8, 9 and 25.

SOLUTION

(i) $12 = 2^2 \times 3$, $15 = 3 \times 5$, $21 = 3 \times 7$. The only common prime is 3, so HCF $= 3$. LCM $= 2^2 \times 3 \times 5 \times 7 = 420$.

(ii) 17, 23 and 29 are all primes with no common factor, so HCF $= 1$. LCM $= 17 \times 23 \times 29 = 11339$.

(iii) $8 = 2^3$, $9 = 3^2$, $25 = 5^2$. There is no common prime, so HCF $= 1$. LCM $= 2^3 \times 3^2 \times 5^2 = 1800$.

Answer: (i) HCF 3, LCM 420 (ii) HCF 1, LCM 11339 (iii) HCF 1, LCM 1800

Q4 Given that HCF $(306, 657) = 9$, find LCM $(306, 657)$.

SOLUTION

For two numbers, HCF $\times$ LCM $=$ product of the numbers.

$$\text{LCM} = \frac{306 \times 657}{9} = 34 \times 657 = 22338.$$

Answer: 22338

Q5 Is it possible for 6^n to end in the digit 0 for some natural number n?

SOLUTION

A number that ends in 0 is divisible by 10, so its prime factorisation must contain both 2 and 5.

$6^n = (2 \times 3)^n = 2^n \times 3^n$. Its only prime factors are 2 and 3.

By the Fundamental Theorem of Arithmetic this factorisation is unique, so 5 can never be a factor of 6^n.

Answer: no, 6^n cannot end with the digit 0

Q6 Explain why $7 \times 11 \times 13 + 13$ and $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ are composite numbers.

SOLUTION

A composite number has a factor other than 1 and itself.

$7 \times 11 \times 13 + 13 = 13\,(7 \times 11 + 1) = 13 \times 78$. It has the factors 13 and 78, so it is composite.

$7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5 = 5\,(7 \times 6 \times 4 \times 3 \times 2 \times 1 + 1) = 5 \times 1009$. It has the factors 5 and 1009, so it is composite.

Answer: each number can be written as a product of two factors greater than 1, so both are composite

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Q7 Two people drive round a circular path. One takes 18 minutes for a round and the other takes 12 minutes. They start together from the same point in the same direction. After how many minutes will they meet again at the starting point?

SOLUTION

They are both at the starting point after a time that is a multiple of 18 and also of 12. The first such time is the LCM of 18 and 12.

$18 = 2 \times 3^2$ and $12 = 2^2 \times 3$, so LCM $= 2^2 \times 3^2 = 36$.

Answer: after 36 minutes

2 Exercise 1.2

Q1 Prove that $\sqrt{5}$ is irrational.

SOLUTION

Suppose $\sqrt{5}$ is rational. Then $\sqrt{5} = \frac{a}{b}$, where a and b are co-prime integers and $b \neq 0$.

Squaring, $a^2 = 5b^2$. So 5 divides a^2, and therefore 5 divides a (if a prime divides a^2, it divides a).

Write $a = 5c$.

Then $25c^2 = 5b^2$, so $b^2 = 5c^2$. So 5 divides b^2, and therefore 5 divides b.

So a and b have the common factor 5, which contradicts the fact that they are co-prime.

The assumption is wrong, so $\sqrt{5}$ is irrational.

Q2 Prove that $3 + 2\sqrt{5}$ is irrational.

SOLUTION

Suppose $3 + 2\sqrt{5}$ is rational. Then $3 + 2\sqrt{5} = \frac{a}{b}$, where a and b are integers and $b \neq 0$.

Rearranging, $\sqrt{5} = \frac{a - 3b}{2b}$.

Since a and b are integers, $\frac{a - 3b}{2b}$ is rational, so $\sqrt{5}$ would be rational. This contradicts the fact that $\sqrt{5}$ is irrational.

The assumption is wrong, so $3 + 2\sqrt{5}$ is irrational.

Q3 Prove that the following are irrational: (i) $\frac{1}{\sqrt{2}}$ (ii) $7\sqrt{5}$ (iii) $6+\sqrt{2}$.

SOLUTION

In each part, suppose the number is rational and equal to $\frac{a}{b}$, where a and b are integers and $b \neq 0$.

(i) $\frac{1}{\sqrt{2}} = \frac{a}{b}$ gives $\sqrt{2} = \frac{b}{a}$ (here $a \neq 0$, as the number is not zero). The right side is rational but $\sqrt{2}$ is irrational, a contradiction.

(ii) $7\sqrt{5} = \frac{a}{b}$ gives $\sqrt{5} = \frac{a}{7b}$. The right side is rational but $\sqrt{5}$ is irrational, a contradiction.

(iii) $6+\sqrt{2} = \frac{a}{b}$ gives $\sqrt{2} = \frac{a-6b}{b}$. The right side is rational but $\sqrt{2}$ is irrational, a contradiction.

So $\frac{1}{\sqrt{2}}$, $7\sqrt{5}$ and $6+\sqrt{2}$ are all irrational.

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